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Different bulk and active bacterial communities in cryoconite from the margin and interior of the Greenland ice sheet

Marek Stibal,[1,2]* Morten Schostag,[2,3] Karen A. Cameron,[1,2] Lars H. Hansen,[4] David M. Chandler,[5] Jemma L. Wadham[5] and Carsten S. Jacobsen[1,2,6]

[1]*Department of Geochemistry, Geological Survey of Denmark and Greenland (GEUS), Copenhagen, Denmark.*
[2]*Center for Permafrost (CENPERM),* [3]*Department of Biology, and* [6]*Department of Plant and Environment, University of Copenhagen, Copenhagen, Denmark.*
[4]*Department of Environmental Science, Aarhus University, Roskilde, Denmark.*
[5]*Bristol Glaciology Centre, School of Geographical Sciences, University of Bristol, Bristol, UK.*

Summary

Biological processes in the supraglacial ecosystem, including cryoconite, contribute to nutrient cycling within the cryosphere and may affect surface melting, yet little is known of the diversity of the active microbes in these environments. We examined the bacterial abundance and community composition of cryoconite over a melt season at two contrasting sites at the margin and in the interior of the Greenland ice sheet, using sequence analysis and quantitative polymerase chain reaction of coextracted 16S rDNA and rRNA. Significant differences were found between bulk (rDNA) and potentially active (rRNA) communities, and between communities sampled from the two sites. Higher concentrations of rRNA than rDNA were detected at the interior site, whereas at the margin several orders of magnitude less rRNA was found compared with rDNA, which may be explained by a lower proportion of active bacteria at the margin site. The rRNA communities at both sites were dominated by a few taxa of Cyanobacteria and Alpha- and/or Betaproteobacteria. The bulk alpha diversity was higher in the margin site community, suggesting that local sources may be contributing towards the gene pool in addition to long distance transport.

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Introduction

Glaciers and ice sheets are considered the coldest terrestrial biome, containing distinct ecosystems with simple food webs that are dominated by microbes (Hodson *et al.*, 2008; Anesio and Laybourn-Parry, 2012). Biological processes in the supraglacial ecosystem, including cryoconite (surface aggregates of mineral and organic particles and microbial cells), contribute to nutrient cycling within the cryosphere and may affect surface melting (Stibal *et al.*, 2012a). Changes in biological process rates have been associated with changes in diversity (Allison and Martiny, 2008); however, while many studies have measured supraglacial biological process rates (Stibal *et al.*, 2008; 2012b; Anesio *et al.*, 2009; 2010; Hodson *et al.*, 2010a,b; Telling *et al.*, 2010; 2012), there is currently little knowledge of the diversity of active microorganisms in these environments.

Microbial communities in cryoconite are dominated by bacteria from the classes Proteobacteria, Cyanobacteria, Bacteroidetes and Actinobacteria, as shown in studies based on molecular analysis of DNA extracted from environmental samples (Xiang *et al.*, 2009; Segawa *et al.*, 2010; Edwards *et al.*, 2011; 2014; Cameron *et al.*, 2012; Zarsky *et al.*, 2013). The relatively low rates of biological processes on glacial surfaces (Hodson *et al.*, 2010a; Telling *et al.*, 2010) are suggestive that only a small proportion of the total diversity within these environments contribute to biological activity. Previously, co-analysis of 16S rRNA and rDNA has been used to address differences between bulk and active bacterial communities in a number of environments, including Alpine glacial streams, Arctic soils and subglacial melt waters (Männistö *et al.*, 2013; Dieser *et al.*, 2014; Wilhelm *et al.*, 2014). Despite the limitations of this technique, it is a valuable tool for the investigation of microbial community dynamics (Blazewicz *et al.*, 2013).

The Greenland ice sheet (GrIS) is the largest body of ice in the northern hemisphere (~ 1.75×10^6 km^2). Its biologically active area, where melt occurs on an annual basis and where liquid water, necessary for biological activity, is generated, has been modelled to typically cover

~ 200 000 km^2 around the margins of the GrIS (Hodson *et al.*, 2010a), and can be up to ~ 100 km wide. Microbial abundance and activity varies considerably within the biologically active region of the GrIS and has been shown to be dependent on the distance from the ice-free land (Stibal *et al.*, 2012b; Telling *et al.*, 2012). This is probably due to differences in environmental disturbances, such as melt water flow (Chu, 2014), sources of microbial inocula and nutrients (Telling *et al.*, 2012; Cameron *et al.*, 2014), and melt season duration (Hodson *et al.*, 2010a; Stibal *et al.*, 2010).

Here we examine for the first time the bacterial abundance, diversity and community composition of cryoconite sampled from two locations on the GrIS – one at the margin and one in the interior of the southwestern part of the ice sheet (~ 30 km inland) – over the course of an entire melt season. We hypothesize that the bulk (rDNA-based) diversity of bacteria is higher at the margin site due to cell input from adjacent ice-free areas, but that the proportion of potentially active (rRNA-based) bacteria is higher at the interior site where higher activity has been measured in previous studies. In order to test this hypothesis, we collected cryoconite at both sites, coextracted DNA and RNA from the samples, and performed quantitative polymerase chain reaction (qPCR) and pyrosequencing of 16S ribosomal DNA and cDNA to determine the abundance, diversity and community composition. Multivariate statistics were then used to identify significant differences between the bulk and active communities, and between communities sampled from the two distinct sites.

Results and discussion

We collected cryoconite samples at two distinct sites on the GrIS over the course of the melt season: on seven occasions (May–September 2012) at an ice marginal site (67°09.1′ N, 50°02.5′ W; 590 m a.s.l.) and on three occasions (June–August 2012) at an interior site (66°57.8′ N, 49°16.5′ W; 1020 m a.s.l.) (Table S1 and Appendix S1). Our data go beyond the 'snapshot' limitation of previous work, and for the first time enable a robust comparison of the bulk and potentially active communities at two ice sheet sites previously shown to be very different in terms of microbial abundance and activity (Stibal *et al.*, 2010; 2012b).

DNA/RNA abundance

At the margin site, 16S rDNA copy numbers per gram of wet weight cryoconite were significantly different (pairwise *t*-test, $t = 17.3$, $P < 0.001$) from, and several orders of magnitude higher, than 16S rRNA ($1.8–94 \times 10^8$ gene copies g^{-1} and $2.6–97 \times 10^4$ gene copies g^{-1} respectively;

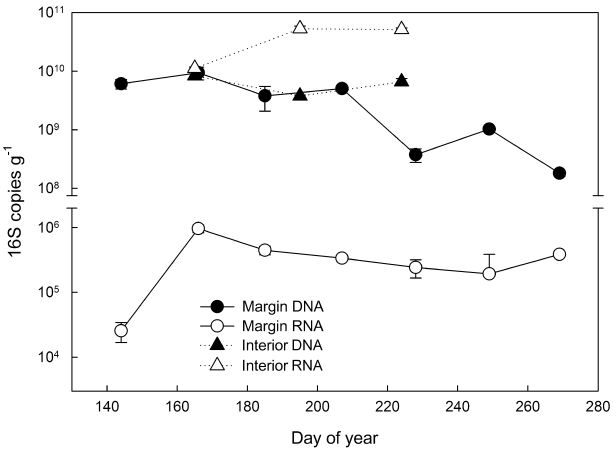

Fig. 1. Abundance of the 16S rRNA genes and 16S rRNA in cryoconite (in copies per gram of wet weight) from the margin and interior sites over the course of the melt season (mean ± SD, $n = 3$). DNA and RNA were coextracted from cryoconite samples preserved by LifeGuard (MO BIO) using the RNA PowerSoil Total RNA Isolation Kit with the RNA PowerSoil DNA Elution Accessory Kit (MO BIO). Between 1.7 and 2.5 g (wet weight) of sediment was used for each extraction. All RNA extracts were immediately subjected to Dnase treatment using the RTS DNase Kit (MO BIO). A subsample of each DNase-treated RNA sample was then used as template in cDNA production via RT-PCR using the RevertAid Premium RT kit and random hexamer primers (Fermentas). Quantitative PCR was performed with primers targeting the bacterial 16S rRNA gene (341F: 5′-CCTAYGGGRBGCASCAG-3′; 806R: 5′-GGACTACNNGGGTATCTAAT-3′).

Fig. 1). Marginal site rDNA copy numbers displayed a decreasing trend over time, whereas rRNA copy numbers showed an initial increase in abundance, after which they remained at a concentration of ~ 10^5 g^{-1} (Fig. 1). The DNA : RNA copy number ratios were between ~ 240 000 on 23 May and ~ 470 on 25 September. Interior site rDNA copy numbers were higher than those at the margin, albeit of the same order of magnitude ($38–83 \times 10^8$ g^{-1}), and significantly different (two-sample *t*-test with unequal variance, $t = 1.83$, $P = 0.040$). The interior site rRNA copy numbers ($110–530 \times 10^8$ g^{-1}) were significantly different from and higher than both marginal site rRNA copy numbers (two-sample *t*-test with unequal variance, $t = 29.6$, $P < 0.001$) and rDNA copy numbers (pairwise *t*-test, $t = 4.70$, $P < 0.001$). The DNA : RNA ratios at the interior site ranged between 0.07 and 0.70. Total organic carbon (TOC) and total nitrogen (TN) contents of the cryoconite were lower at the margin (0.44–1.1 and 0.27–0.41 mg g^{-1}, respectively) compared with the interior (29–30 and 5.2–5.4 mg g^{-1}, respectively), consistent with previous studies (Stibal *et al.*, 2010; 2012b; Telling *et al.*, 2012). The TOC : TN molar ratios were between 1.7–4.0 at the margin and 6.4–6.5 at the interior site (Fig. S1).

Our qPCR results (10^8–10^9 16S rDNA copies g^{-1}) are comparable to microscopy enumerations from previous studies of the same area (Stibal *et al.*, 2012b; Bellas

et al., 2013) and to enumerations made at other supraglacial sites (Stibal *et al.*, 2008; Hodson *et al.*, 2010a,b; Langford *et al.*, 2010), assuming that prokaryotic cells usually contains 1–15 copies of the 16S rRNA gene (Klappenbach *et al.*, 2001). The higher rDNA gene copy numbers found at the interior site compared with the margin site are consistent with more cells found previously in the interior (Stibal *et al.*, 2012b). However, 16S rRNA abundances and RNA : DNA ratios varied considerably between the margin and the interior. More rRNA than rDNA was detected at the interior site, whereas at the margin several orders of magnitude less rRNA (2.6–97 $\times 10^4$ copies g^{-1}) was found compared with rDNA (1.8–94 $\times 10^8$ copies g^{-1}). This difference may be explained by a lower proportion of active bacteria at the margin site. These results differ from those from a Canadian valley glacier, likely analogous to our margin site due to its proximity to terrestrial sources of cells and nutrients, where the DNA : RNA ratios in cryoconite were found to be between 2 and 3 (Hamilton *et al.*, 2013). Despite this, we deduce that our results are not a methodological artifact, since samples from different dates were handled separately, and the margin samples were processed and analysed in parallel with samples from the interior site where no such difference between rRNA and rDNA was found.

Bulk and active diversity

Table 1 shows the results of 454 pyrosequencing of the 16S rRNA gene and 16S rRNA cDNA from the two sites, including the richness and the alpha diversity calculated as the Shannon–Wiener and Chao1 indices. The average sequence length after quality check was 441 bp (ranging between 173 and 493 bp). Between 586 and 7498 sequences were obtained for each sample after filtering, denoising and chimera check (see Appendix S1 for method details). There was a significant difference in alpha diversity between the DNA and cDNA amplicon libraries at the margin site (pairwise *t*-test; Shannon–Wiener: $t = 2.33$, $P = 0.029$; Chao1: $t = 3.76$, $P = 0.005$); however, no significant differences in alpha diversity were found between DNA and cDNA at the interior site (Shannon–Wiener: $t = 1.30$, $P = 0.16$; Chao1: $t = 0.56$, $P = 0.31$). The DNA diversity was higher at the margin site compared with the interior (two-sample *t*-test with unequal variance; Shannon–Wiener: $t = 3.17$, $P = 0.012$; Chao1: $t = 2.71$, $P = 0.027$), while in contrast the Chao1 index for RNA diversity was higher at the interior site in comparison to the marginal site ($t = 2.04$, $P = 0.044$). No significant difference in the Shannon–Wiener index for the RNA diversity was found between the sites ($t = 0.07$, $P = 0.47$).

Differences in community composition were found between the margin and the interior site, and between the bulk (rDNA) and potentially active (rRNA) community, as visualized by a principal component analysis of the abundance-normalized dataset (Fig. 2). The first principal component explained 31% and the second principal component 20% of the variation within the data; 67% of variation was explained by the first four components. The significance of the differences was tested using two separate redundancy analyses (RDA) with either DNA/RNA or site as the explanatory variable, and the other as a covariable respectively (Appendix S1). Both factors were found significant (DNA/RNA explained 13.4% of variation in the data, $F = 5.9$, $P = 0.002$; site explained 23.9%, $F = 5.1$, $P = 0.001$). The RDAs were then repeated using the qPCR-based abundance data as an additional

Table 1. Results of 454 pyrosequencing of rDNA/rRNA (466 bp amplicons flanking the V3 and V4 regions of the 16S rRNA gene amplified using the primers 341F and 806R) from the margin and interior sites collected over the course of the melting season 2012.

		DNA			RNA		
			Diversity			Diversity	
Site	Date (DOY)	Richness	S-W	Chao1	Richness	S-W	Chao1
Margin	23 May (144)	83	3.95	148	70	4.13	111
	14 June (166)	107	5.01	199	58	4.30	74
	3 July (185)	121	5.14	214	43	2.90	54
	25 July (207)	114	5.33	200	11	0.92	21
	15 August (228)	109	5.15	209	91	4.93	158
	5 September (249)	113	5.19	192	68	4.06	110
	25 September (269)	77	4.92	117	70	3.94	108
Interior	13 June (165)	87	4.51	157	57	3.13	106
	13 July (195)	71	3.92	104	82	3.81	165
	11 August (224)	67	3.88	104	77	3.74	154

The alpha diversity indices, Shannon–Wiener (S–W) and Chao1 were calculated from rarefied and re-sampled data and are presented as averages. The 16S rRNA and rDNA sequences have been deposited in the Sequence Read Archive (SRA) at the National Center for Biotechnology Information (NCBI) under Accession Number SRP035650.
DOY, day of year.

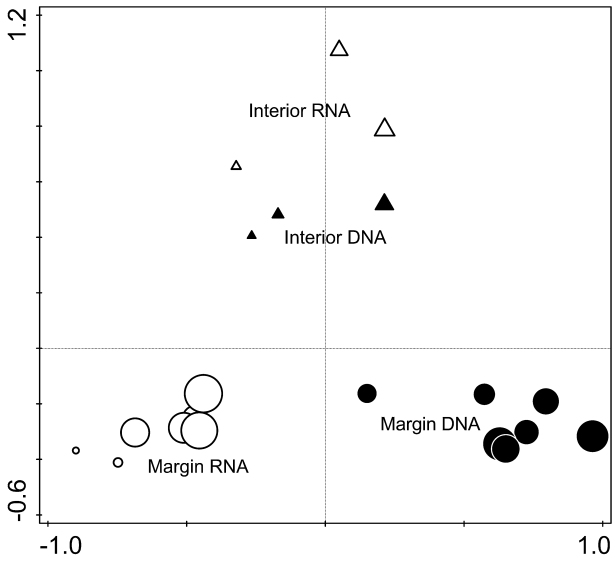

Fig. 2. Principal component analysis of the bulk and potentially active community composition data from margin and interior sites on the GrIS. The sequencing data were rarefied and randomly re-sampled 1000 times to the lowest number of sequences per sample (586 reads). Average values from the 1000 re-samplings were then used for further analysis. The relative community composition data were abundance-normalized using the qPCR data shown in Fig. 1, ln(x+1) transformed, standardized and centred prior to analysis with CANOCO 5. The size of symbols is proportional to sample richness. Full circles denote margin DNA, empty circles margin RNA, full triangles interior DNA and empty triangles interior RNA.

covariate in order to account for its large variation, with similar results (DNA/RNA explained 13.4% of variation in the data, $F = 2.5$, $P = 0.028$; site explained 24.5%, $F = 5.2$, $P = 0.002$).

Amplicons relating to 10 bacterial phyla were found to contribute towards at least 1% of sequences on at least one date and/or site (Fig. S2). The bulk community at the margin site was dominated by Cyanobacteria, whose relative abundance decreased over the course of the season from ~69% to 23% of sequences (Fig. 3). Other well-represented groups in the bulk community included Proteobacteria, whose abundance increased from 7.9% in May to 37% in September, Actinobacteria (relative abundance: 11–32%) and Bacteroidetes (6.7–24%). The potentially active community was dominated by Proteobacteria (44–99%), followed by Cyanobacteria (up to 25% of sequences), Actinobacteria (0.3–11%) and Bacteroidetes (0.1–11%) (Fig. 3 and Fig. S2). The interior site bulk community was dominated by Cyanobacteria (19–32% of sequences), Chloroflexi (16–29%), Bacteroidetes (16–29%), Proteobacteria (7.6–15%) and the candidate phylum WPS-2 (8.4–10%). The potentially active community was dominated by Proteobacteria (38–65%), Cyanobacteria (20–46%) and WPS-2 (8.9–25%) (Fig. 3). The most represented taxa in the RNA from both sites are listed in Table S1. Our results are in agreement with recent data from the same area (Edwards *et al.*, 2014), in which Proteobacteria were found (co)dominant in the bulk community, and more Alphaproteobacteria were found in the interior relative to Betaproteobacteria.

Figure 4 shows a comparison of relative abundances of the most represented phyla in the bulk and the potentially active community. Alphaproteobacteria were more represented in the RNA amplicon library compared with the DNA at both the margin and the interior sites (pairwise *t*-test; margin: $t = 2.32$, $P = 0.029$; interior: $t = 5.23$, $P = 0.017$), while Beta- and Gammaproteobacteria were more represented in the RNA only at the margin (Beta: $t = 2.22$, $P = 0.034$; Gamma: $t = 7.07$, $P < 0.001$). In comparison to DNA amplicon libraries, Actinobacteria (margin: $t = 4.01$, $P = 0.004$; interior: $t = 3.63$, $P = 0.034$), Bacteroidetes (margin: $t = 3.31$, $P = 0.008$; interior: $t = 5.47$, $P = 0.016$) and the candidate phylum TM7 (margin: $t = 3.01$, $P = 0.012$; interior: $t = 10.3$, $P = 0.005$) were less represented in the RNA amplicon library at both sites, while Cyanobacteria were less represented in the RNA library only at the margin ($t = 4.18$, $P = 0.003$). Comparisons of sites revealed that samples from the margin site contained a higher proportion of Betaproteobacteria (two-sample *t*-test with unequal variance; DNA: $t = 4.82$, $P = 0.001$; RNA: $t = 3.83$, $P = 0.004$), Gammaproteobacteria (RNA only, $t = 7.98$, $P < 0.001$), Actinobacteria (DNA: $t = 5.21$, $P < 0.001$; RNA: $t = 4.56$, $P = 0.002$), Firmicutes (DNA: $t = 2.20$, $P = 0.029$; RNA: $t = 2.08$, $P = 0.041$), Thermi (DNA: $t = 4.48$, $P = 0.002$; RNA: $t = 2.96$, $P = 0.013$) and TM7 (DNA: $t = 4.31$, $P = 0.003$; RNA: $t = 2.65$, $P = 0.019$), while the interior site contained more Cyanobacteria (RNA only; $t = 2.53$, $P = 0.043$), WPS-2 (DNA only: $t = 12.2$, $P < 0.001$) and Acidobacteria (RNA only: $t = 2.84$, $P = 0.018$).

Diversity sources

All the dominant taxa within the potentially active community at the interior site were most closely related to amplicons generated from glaciers or similar cold environments (Table S1; Pointing *et al.*, 2009; Xiang *et al.*, 2009; Segawa *et al.*, 2010; Chong *et al.*, 2012; Zeng *et al.*, 2013), whereas only some of the dominant taxa from the margin site samples were similar to sequences generated from cold environments (Table S1; Pradhan *et al.*, 2010). These results may be explained by differences in sources and sinks of bacterial cells and their activity between the margin and the interior of the ice sheet. Bacterial growth may explain the higher abundance of DNA, and in particular, of RNA, at the interior site. This is supported by *in situ* measurements of microbial production from the same transect (Stibal *et al.*, 2012b) and also by the higher content of organic carbon at the interior site (Fig. S1; Stibal

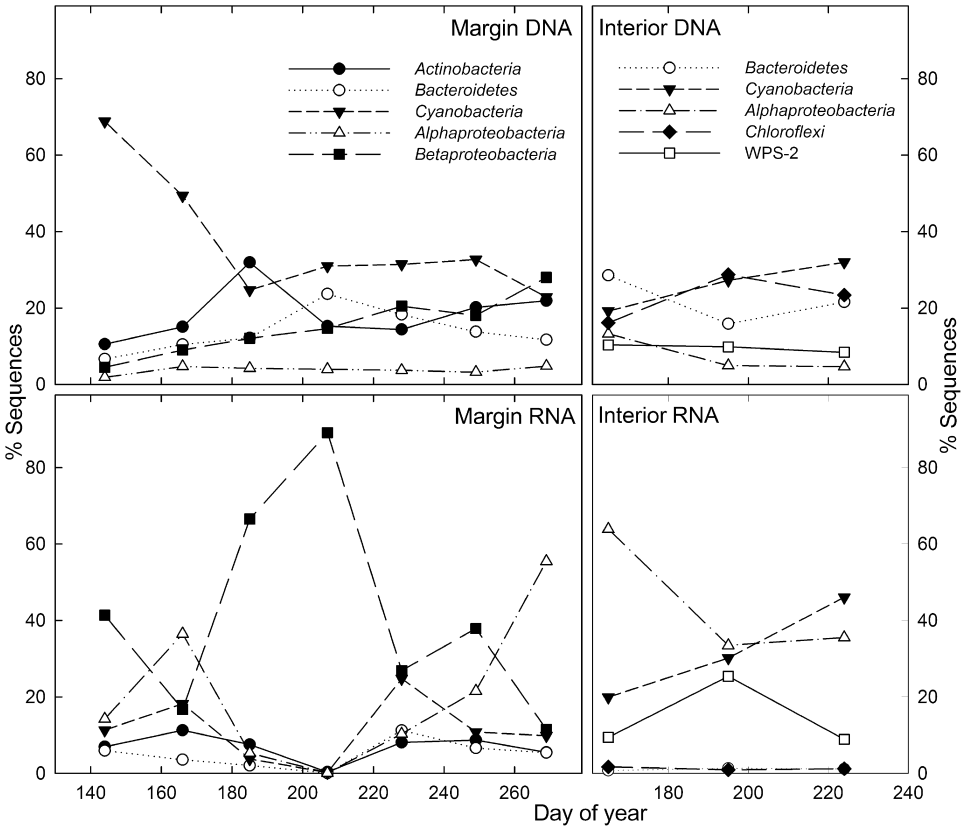

Fig. 3. Relative abundance of major phyla (> 3%) in the bulk and potentially active communities from the margin and interior sites over the course of the 2012 melt season.

et al., 2010; 2012b), possibly originating from primary production, consistent with the higher proportion of Cyanobacteria (Fig. 3). The higher DNA diversity at the margin suggests a local input of bacterial cells, probably from adjacent ice-free environments, such as moraines, soils, lakes and wetlands, as observed elsewhere (Harding *et al.*, 2011; Edwards *et al.*, 2013; Cameron *et al.*, 2014), since similar diversity and community composition would be expected for the bulk community at both sites if atmospheric deposition of cells from a global pool of cold-tolerant and/or cold-loving microbes transported over large distances (Jungblut *et al.*, 2010; King *et al.*, 2010) was the

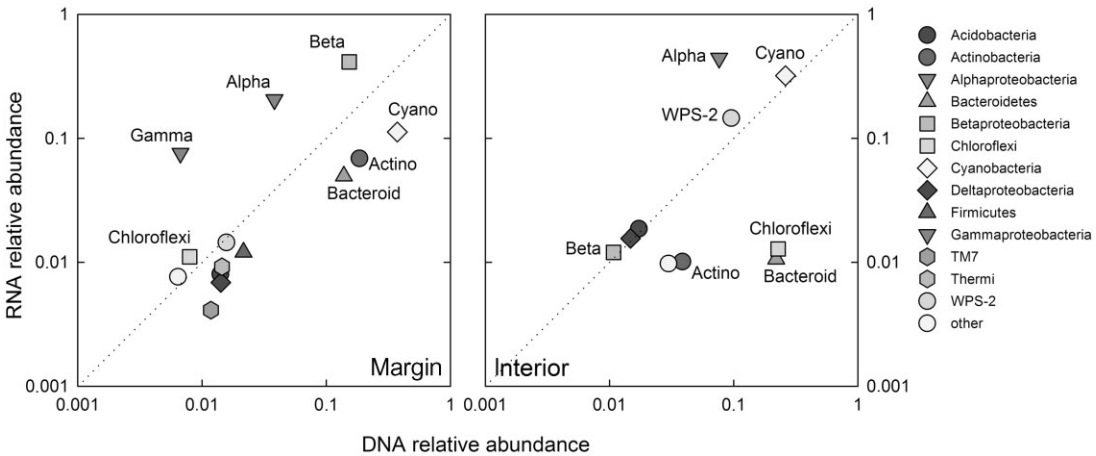

Fig. 4. Relative abundance of phyla in the bulk and potentially active communities from the margin and interior sites. The abundance shown was averaged over time for both sites. Note the logarithmic scales of the axes.

only source for the GrIS. Further analysis of airborne microbes that are deposited on the GrIS is essential to determine the abundance and diversity of GrIS microbial sources, similar to studies performed in Svalbard (Irvine-Fynn *et al*., 2012). In addition, production of exopolysaccharides by, for example, Cyanobacteria (Hodson *et al*., 2010b; Langford *et al*., 2010) or Betaproteobacteria (Besemer *et al*., 2012), may create a selective advantage as it can facilitate resistance to washout where surface slope allows for rapid flow (Chu, 2014).

Community structure

Despite the difficulty of inferring ecosystem functions from diversity, the simplicity of the active community structure in cryoconite on the GrIS allows us to draw a crude picture. The potentially active community at both sites was made up mainly by a few taxa of Cyanobacteria and Alpha- and/or Betaproteobacteria. This supports the notion that cryoconite holes are simple environments, with Cyanobacteria as primary producers and various heterotrophic bacteria, mostly Proteobacteria, as decomposers. The dominant taxa of the potentially active primary producers in the community (*Leptolyngbya*, *Pseudanabaena*, *Nostoc*; Table S1) have been previously described from glacier surfaces (Stibal *et al*., 2006; Xiang *et al*., 2009; Segawa and Takeuchi, 2010). It is likely that a combination of easy transportability and a tolerance to freezing/desiccation stress and low nutrient concentrations (Vincent, 2000; Tamaru *et al*., 2005; Jungblut *et al*., 2010) facilitates the viability of these species within the supraglacial environment. In addition, ability to produce exopolysaccharides may contribute towards a freezing and desiccation tolerance (Tamaru *et al*., 2005) and may play a role in resisting melt water flow. The dominant taxa of potentially active decomposers belonged to several Proteobacterial families. Members of the Comamonadaceae (Betaproteobacteria), which were well represented in our samples (e.g. *Delftia*), have been known to successfully colonize cold and nutrient-deficient environments including glaciers, and their success has been attributed to their ability to survive low temperature and long-range transport, as well as to utilize a broad range of organic substrates (Darcy *et al*., 2011; Franzetti *et al*., 2013). Sphingomonadaceae (Alphaproteobacteria), a member of which was dominant at the interior site, are also known from glacial environments (Simon *et al*., 2009; Edwards *et al*., 2011), and members of this family have additionally been shown to degrade a wide range of organic carbon substrates, including recalcitrant compounds and pollutants, at low nutrient concentrations (White *et al*., 1996). An exception from the Proteobacterial dominance of decomposers was an OTU of the candidate phylum WPS-2. Although it is similar to clones from other Arctic glaciers (Segawa *et al*., 2010; Zeng *et al*., 2013) and so unlikely to be a contamination, this group was described from polluted temperate soil (Nogales *et al*., 2001) and its metabolism is unknown. Bacteroidetes and Actinobacteria, often found to be dominant in DNA extracted from glacial environmental samples (Simon *et al*., 2009; Xiang *et al*., 2009; Segawa *et al*., 2010; Edwards *et al*., 2011; Cameron *et al*., 2012; Zarsky *et al*., 2013), were less represented in the RNA compared with the DNA in our samples, suggesting that these groups may be constantly supplied from exogenous sources and/or that they remain inactive on the ice surface for long periods of time.

Due to time constraints and the necessity to pool cryoconite from a large area of the ice surface (~ 25 m^2) in order to acquire enough material for chemical and microbiological analysis, we could not analyse replicate samples from individual sampling days. Consequentially, we cannot draw any conclusions about seasonal change in the bacterial community. We are also aware of the limitations of using rRNA as an indicator of current microbial activity (Blazewicz *et al*., 2013). Specifically, the numbers of rDNA and rRNA copies can vary among taxa (Moeseneder *et al*., 2005), and dormant cells can also contain high numbers of ribosomes (Sukenik *et al*., 2012). Therefore, we eschew direct comparison of microbial groups within the communities, and acknowledge that potentially active communities may include dormant cells.

In summary, our results show for the first time significant differences between the bulk and potentially active bacterial communities in cryoconite collected at two different sites at the margin and the interior of the GrIS. Higher copy numbers of rRNA than rDNA were detected at the interior site, whereas at the margin several orders of magnitude less rRNA were detected compared with rDNA, which may be explained by a lower proportion of active bacteria at the margin site. The higher DNA diversity at the margin suggests that localized inputs of bacterial cells may factor alongside sources originating from long-distance atmospheric transportation. The higher abundance of DNA, and in particular RNA, at the interior site compared with the margin may be the result of elevated bacterial growth, which is supported by previous productivity measurements from the same area. The potentially active communities at both sites were predominantly composed of Cyanobacteria and Alpha- and/or Betaproteobacteria, suggesting that GrIS cryoconite is a simple environment with a few taxa of photoautotrophic Cyanobacteria and a limited diversity of heterotrophic Proteobacteria.

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Supporting information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Fig. S1. Total organic carbon (TOC) and total nitrogen (TN) contents in cryoconite (in mg g^{-1} of dry weight) from the margin and interior sites over the course of the melt season.
Fig. S2. Phylum level relative abundance of rDNA and rRNA from margin and interior sites. Only phyla contributing towards over 1% of abundance on at least one sample date are shown; the rest are compressed into 'other'.
Table S1. Dominant taxa (> 3%) in the active (RNA) communities from the margin and interior sites on the GrIS. α = *Alphaproteobacteria*, β = *Betaproteobacteria*, γ = *Gammaproteobacteria*.
Appendix S1. Supplementary methods.